UCS503P — Software Engineering Lab

FoodBridge

A Surplus Food Redistribution Matcher

Software Requirements Specification

Version 1.0

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1 Introduction

1.1 Purpose

This document specifies the software requirements for **FoodBridge**, a web-based system that matches surplus edible food from donor organisations to nearby welfare organisations able to consume it within its usable window. It covers the complete system: the React client, the Express API, the Python matching service and the PostgreSQL/PostGIS database.

The specification describes release 1.0, the prototype demonstrated at the mid-semester evaluation, and identifies requirements deferred to a later iteration. Deferred requirements are marked explicitly rather than omitted, so that the boundary of the delivered system is unambiguous.

1.2 Document Conventions

Functional requirements carry the prefix **FR**, non-functional requirements **NFR** and business rules **BR**. Each requirement is atomic and testable.

Priority is stated as one of:

Essential Required for the system to meet its objectives; implemented in release 1.0.

Desirable Improves the system materially but is not required for operation.

Deferred Agreed as in scope but scheduled for a later iteration.

Identifiers in `monospace` are actual names in the implementation, so a requirement can be traced to code.

1.3 Intended Audience and Reading Suggestions

The document is written for the course instructor and evaluators, the three developers, and any subsequent maintainer.

Readers wanting the problem and its scope should read Sections 1 and 2. Developers implementing or testing a feature should read Section 4, which holds the functional requirements. Evaluators assessing quality attributes should read Section 5. Appendix B contains the analysis models.

1.4 Product Scope

Restaurants, hostel and institutional mess halls, caterers and event organisers routinely generate surplus edible food, while nearby shelters and community kitchens struggle to source food at short notice. The food is safe and immediately consumable, but only within a short and shrinking window.

FoodBridge translates an informal, telephone-and-messaging redistribution practice into a measurable software workflow in which allocation decisions are recorded, reviewable and made against explicit criteria. The binding constraint is time, not willingness: a coordination process that consumes most of a donation's usable window converts a solvable logistics problem into waste.

In scope. Donor submission; administrative review of each donation; automated eligibility filtering and scored allocation; offer, acceptance and decline; a two-stage pickup record; in-app notification; and an impact dashboard computed from system timestamps.

Out of scope. The system coordinates but does not dispatch: physical transport is arranged between the two matched parties. Payment, vehicle routing, cold-chain monitoring and automated verification of an organisation's legal status lie outside the boundary.

1.5 Project Objectives

OBJ-1 Deliver an end-to-end donation workflow — post, review, match, collect, confirm — with the simplest allocation rule that works, so that a working system exists from the first iteration.

OBJ-2 Reduce the median Time-to-Match (TTM) between donation submission and an accepted match to 20 minutes or less, measured purely from database timestamps rather than self-report.

OBJ-3 Make every allocation decision explainable and improvable by isolating the matching logic behind a stable service interface, first as a greedy best-fit assignment and later as a global assignment using the Hungarian method, without disrupting the rest of the system.

1.6 References

1. FoodBridge Project Proposal, UCS503P Group 3, August 2026.
2. IEEE Std 830-1998, *IEEE Recommended Practice for Software Requirements Specifications*.
3. H. W. Kuhn, "The Hungarian Method for the Assignment Problem," *Naval Research Logistics Quarterly*, vol. 2, pp. 83–97, 1955.
4. PostGIS Project Steering Committee, *PostGIS Manual*, 2023.

2 Overall Description

2.1 Product Perspective

FoodBridge is a new, self-contained system with no predecessor. It is built as three cooperating processes over one database, so that the allocation logic can be replaced without touching the application.

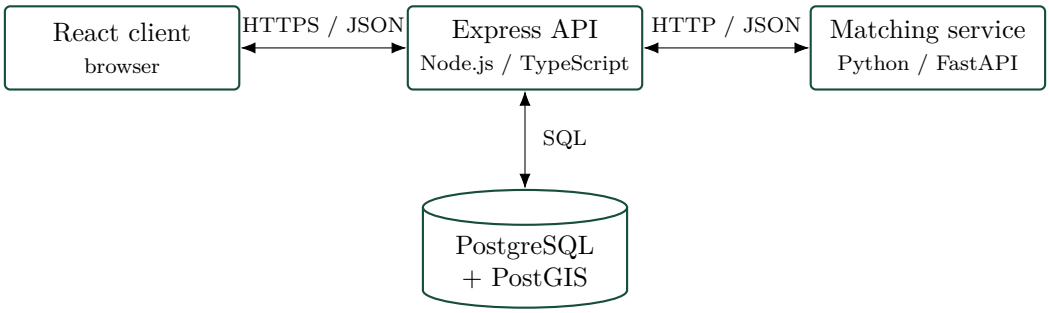


Figure 1: System architecture. The matching service holds no state; all persistence is the API's responsibility.

2.2 Product Functions

- Register and authenticate donors, recipient organisations and administrators.
- Accept a surplus food listing with quantity, type, location and expiry deadline.
- Allow an administrator to approve or reject each listing before it is matched.
- Filter organisations on hard constraints and rank the remainder on a weighted score.
- Offer a donation to the best-fit organisation and re-offer on decline or timeout.
- Record a two-stage pickup, confirmed by the collecting organisation.
- Notify both parties at every transition.
- Report impact metrics derived from system timestamps.

2.3 User Classes and Characteristics

Donor (`DONOR`)
A mess hall, canteen, restaurant, caterer or event organiser. Technically unsophisticated, often submitting from a mobile device while clearing a kitchen. Interacts briefly and infrequently. The dominant class by volume of interactions; the submission form is designed around it.

Recipient Organisation (`RECIPIENT`)
A shelter, NGO, community kitchen, orphanage, old-age home or food bank. Represented by a coordinator who must answer an offer within a bounded window. Interacts more frequently than a donor and with more at stake, since an unanswered offer holds capacity.

Administrator (`ADMIN`)
A member of the operating team. Reviews every submitted donation, triggers matching cycles and monitors impact. Technically capable. Sits on the critical path: no donation is matched without an administrator.

2.4 Stakeholders

Stakeholder	Interest in the system
Donor organisations	Dispose of surplus responsibly with a record that it was collected; minimal effort per donation.
Recipient organisations	Source food at short notice matching their dietary constraints and capacity.
Beneficiaries	Indirect: receive redistributed meals. Not system users.
Operating team	Keep unsuitable listings out and see whether the system is working.
Development team	Deliver a correct, demonstrable and measurable system within the course timeline.
Course instructor	Assess engineering process, correctness and measurement against the proposal.

2.5 Operating Environment

- **Client:** any modern browser on desktop or mobile. No installation.
- **API:** Node.js 20 or later, Express, TypeScript.
- **Matching service:** Python 3.9 or later, FastAPI, SciPy, NumPy.

- **Database:** PostgreSQL 17 with the PostGIS 3.x spatial extension.
- **Network:** the three processes communicate over HTTP; only the client requires public reachability in deployment.

2.6 Design and Implementation Constraints

CON-1 The interface must remain usable on low-to-mid range mobile devices. A donor's entire obligation must fit in one short form.

CON-2 Allocation must use classical, library-backed methods. No novel or learned algorithm may be introduced; the Hungarian method is consumed as a standard library primitive.

CON-3 The matching service must remain stateless and free of database dependencies, so it can be tested in isolation and replaced behind a fixed interface.

CON-4 Evaluation metrics must be derived from database timestamps written by the system, never from values entered by a user.

CON-5 Transport is arranged by the two matched parties. The system must not model vehicles, routes or drivers.

CON-6 Deployment must rely only on common web hosting, a managed relational database with a spatial extension, and free-tier external services.

CON-7 Geocoding must use a free-tier provider, and results must be cached to respect its rate limit of one request per second.

2.7 Assumptions and Dependencies

- Both parties have a smartphone with intermittent internet access.
- An administrator is available to review submissions during operating hours. The consequence of this assumption for OBJ-2 is discussed in Section 5.1.
- Organisations declare their capacity, service radius and dietary constraints honestly; the system does not verify them independently.
- Food safety remains the donor's legal responsibility. Administrative review is a screening step, not a certification.
- The external geocoding provider remains available and free to use.

3 External Interface Requirements

3.1 User Interfaces

A responsive single-page web application. Navigation is role-aware: each user class is shown only the screens belonging to it, and an attempt to reach another class's screen is refused.

Screen	User class	Purpose
Sign in / Register	All	Authentication; registration branches by role.
Post donation	Donor	Single-screen submission form.
My donations	Donor	Listing with live countdowns; detail shows the status timeline.

Screen	User class	Purpose
Offers	Recipient	Pending offers with score breakdown and response countdown.
Collections	Recipient	Accepted donations; pickup confirmation.
Review queue	Administrator	Approve or reject submissions; trigger a matching cycle; view the waitlist.
Impact	Administrator	Metrics derived from system timestamps.

Minimum touch target 44 px; every state change confirmed with visible feedback; errors reported against the specific field that caused them.

3.2 Hardware Interfaces

None. The system requires no sensors, scanners or dedicated hardware. Location is supplied as coordinates, or as a textual address resolved by the geocoding service.

3.3 Software Interfaces

PostgreSQL / PostGIS
Accessed over the Postgres wire protocol. Spatial predicates (`ST_DWithin`, `ST_Distance`) are evaluated in the database, so candidate retrieval does not degrade linearly with the number of registered organisations.

Matching service
A single operation, `POST /match/run`, accepting open donations, eligible organisations, precomputed pair distances and a configuration block; returning assignments, a waitlist and per-candidate explanations.

Geocoding provider
OpenStreetMap Nominatim over HTTPS, at most one request per second, with all results cached in `geocoding_cache`.

3.4 Communications Interfaces

All interfaces are JSON over HTTP. The client authenticates with a bearer token obtained at sign-in and presented on every subsequent request. Errors use a single envelope so the client can render any failure uniformly:

```
{ "error": { "code": string, "message": string, "details"?: any } }
```

HTTP status codes are used as specified: 400 validation, 401 authentication, 403 authorisation, 404 missing, 409 illegal state transition, 502 matching service unreachable.

4 System Features

Figure 2 shows the actors and their use cases. The functional requirements that follow are grouped by feature.

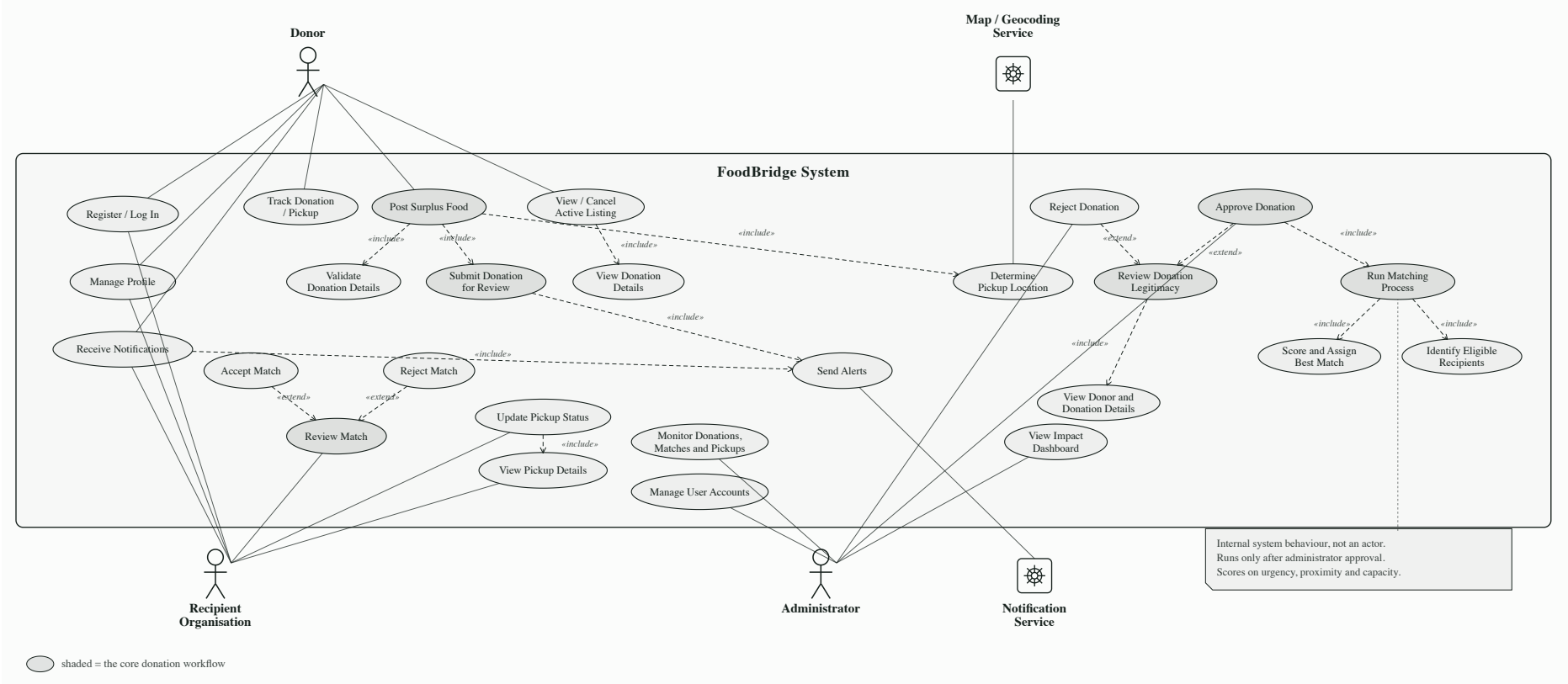


Figure 2: Use case diagram. Dashed arrows are stereotyped dependencies: *include* for behaviour the base use case always performs, *extend* for behaviour that occurs only under a condition.

Table 3 traces every use case in the figure to the requirements that specify it, so that no element of the diagram is left unspecified and no requirement is unmotivated.

Table 3: Use case to requirement traceability.

Use case	**Feature (Section)**	**Requirements**
Register / Log In	Account Management (4.1)	FR-1–FR-6
Manage Profile	Account Management (4.1)	FR-4, FR-7
Post Surplus Food	Donation Submission (4.2)	FR-8, FR-9
Validate Donation Details	Donation Submission (4.2)	FR-10
Determine Pickup Location	Donation Submission (4.2)	FR-11
Submit Donation for Review	Donation Submission (4.2)	FR-12, FR-13
View / Cancel Active Listing	Donation Submission (4.2)	FR-14
View Donation Details	Donation Submission (4.2)	FR-15
Track Donation / Pickup	Donation Submission (4.2)	FR-15, FR-38
Review Donation Legitimacy	Administrative Review (4.3)	FR-16, FR-19, FR-20
View Donor and Donation Details	Administrative Review (4.3)	FR-16
Approve Donation	Administrative Review (4.3)	FR-17, FR-21
Reject Donation	Administrative Review (4.3)	FR-18, FR-21
Run Matching Process	Matching and Allocation (4.4)	FR-26, FR-27, FR-31
Identify Eligible Recipients	Matching and Allocation (4.4)	FR-22, FR-28, FR-30
Score and Assign Best Match	Matching and Allocation (4.4)	FR-23–FR-25, FR-29
Review Match	Offer Response (4.5)	FR-32, FR-35
Accept Match	Offer Response (4.5)	FR-33, FR-36, FR-37
Reject Match	Offer Response (4.5)	FR-34, FR-36
Update Pickup Status	Pickup and Completion (4.6)	FR-39–FR-41
View Pickup Details	Pickup and Completion (4.6)	FR-38
Send Alerts	Notification (4.7)	FR-42, FR-44
Receive Notifications	Notification (4.7)	FR-43
View Impact Dashboard	Impact Reporting (4.8)	FR-45–FR-49
Monitor Donations, Matches and Pickups	Administrative Oversight (4.11)	FR-54 *(deferred)*
Manage User Accounts	Administrative Oversight (4.11)	FR-55 *(deferred)*

The last two rows are the only use cases in the diagram that release 1.0 does not implement; they are specified in Section 4.11 and marked deferred rather than removed from the diagram, so that the team's agreed scope and the delivered scope can be compared directly.

4.1 Account Management and Authentication

Priority: Essential. Every interaction other than registration and sign-in requires an authenticated principal whose role determines what is permitted.

FR-1 The system shall allow a visitor to register as a donor or as a recipient organisation, capturing name, e-mail, password and role.

FR-2 The system shall reject a password shorter than eight characters.

FR-3 The system shall store passwords only as a one-way hash; no plaintext password shall be persisted or logged.

FR-4 The system shall capture, for a recipient organisation, its organisation type, serving capacity, service radius in kilometres, accepted food types, dietary restrictions and

location.

FR-5 The system shall issue a time-limited bearer token at successful sign-in and reject an expired or malformed token with HTTP 401.

FR-6 The system shall derive the caller's role from the token and refuse, with HTTP 403, any request to an operation that role does not permit.

FR-7 The system shall allow an authenticated user to retrieve and update their own profile, including an organisation's capacity, radius, preferences and active flag.

4.2 Donation Submission

Priority: Essential. The donor's entire obligation.

FR-8 The system shall allow a donor to submit a donation specifying food type, quantity in whole servings, pickup location and an expiry deadline.

FR-9 The system shall accept the expiry deadline either as an absolute timestamp or as a number of hours from submission.

FR-10 The system shall reject a submission whose quantity is not a positive whole number, or whose deadline is not in the future, reporting the offending field.

FR-11 The system shall resolve the pickup location server-side: explicit coordinates are used as given; otherwise a textual address is resolved through the geocoding cache and then, on a miss, the external provider.

FR-12 The system shall assign every new donation the status `PENDING_REVIEW` and record its submission timestamp.

FR-13 The system shall notify all administrators that a submission awaits review.

FR-14 The system shall allow a donor to cancel their own donation while it is `PENDING_REVIEW` or `POSTED`, and not afterwards.

FR-15 The system shall allow a donor to view their own donations and the full history of any one of them, and shall refuse access to another donor's donation.

4.3 Administrative Review

Priority: Essential. No donation reaches an organisation without an approval, which is the system's screening control for listing quality and food safety.

FR-16 The system shall present administrators with a queue of all donations in `PENDING_REVIEW`.

FR-17 The system shall allow an administrator to approve a donation, moving it to `POSTED` and making it eligible for matching.

FR-18 The system shall allow an administrator to reject a donation with recorded remarks, moving it to `REJECTED`.

FR-19 The system shall persist every review decision, its author and its timestamp, as an audit record that survives subsequent decisions.

FR-20 The system shall refuse a second review of an already-reviewed donation with HTTP 409.

FR-21 The system shall notify the donor of the review outcome.

4.4 Matching and Allocation

Priority: Essential. The system's reason for existing, and the component OBJ-3 isolates behind a service boundary.

FR-22 The system shall exclude from consideration, before any scoring, any organisation that is inactive, lacks sufficient remaining capacity for the donation, does not accept the donation's food type, or lies outside the current search radius.

FR-23 The system shall compute, for every surviving organisation, a weighted score over three normalised components — urgency (0.40), proximity (0.35) and capacity fit (0.25).

FR-24 The system shall treat the component weights as configuration rather than as embedded logic.

FR-25 The system shall persist, with every offer, its total score and each component score, so that any allocation can be explained after the fact.

FR-26 The system shall support two allocation strategies — a greedy best-fit assignment and a global assignment using the Hungarian method — selectable per matching cycle.

FR-27 The system shall begin matching at a search radius of 5 km and widen it in 5 km steps to a maximum of 25 km when no organisation is eligible.

FR-28 The system shall place on a waitlist, rather than discard, any donation for which no organisation is eligible at the maximum radius, and shall retry it on each subsequent cycle, recording the number of attempts.

FR-29 The system shall reserve an organisation's capacity for the duration of an outstanding offer, so that the same capacity cannot be promised to two donations simultaneously.

FR-30 The system shall return a per-candidate explanation for every donation processed, stating for each organisation whether it was eligible and, if not, which constraint excluded it.

FR-31 The system shall report an unreachable matching service as HTTP 502 and shall not alter donation state in that event.

4.5 Offer Response

Priority: Essential.

FR-32 The system shall present to a recipient organisation only those offers directed at it.

FR-33 The system shall allow an organisation to accept an offer, moving the offer to `ACCEPTED` and creating a scheduled pickup.

FR-34 The system shall allow an organisation to reject an offer, moving the donation back to `POSTED`, excluding that organisation from the immediate retry, and running a fresh matching cycle.

FR-35 The system shall expire an offer unanswered after its response window of 30 minutes, release the capacity it reserved, and return the donation for re-matching.

FR-36 The system shall refuse a response to an offer that has already been answered or has expired, with HTTP 409.

FR-37 The system shall disclose the counterparty's contact details only after an offer has been accepted.

4.6 Pickup and Completion

Priority: Essential.

FR-38 The system shall create a pickup record in `SCHEDULED` when an offer is accepted, carrying the scheduled time and the contact details.

FR-39 The system shall allow the collecting organisation to record collection, moving the pickup to `PICKED_UP` and the donation to `PICKED_UP` in one transaction.

FR-40 The system shall allow the collecting organisation to confirm receipt, moving the pickup and the donation to `COMPLETED` and releasing the organisation's reserved capacity, in one transaction.

FR-41 The system shall refuse any transition not permitted from the current state, with HTTP 409 and without partial effect.

4.7 Notification

Priority: Essential for the workflow; the channel is in-app only in release 1.0.

FR-42 The system shall raise an in-app notification to the affected parties at every state transition of a donation, offer or pickup.

FR-43 The system shall report an unread count and allow a user to mark one notification, or all of them, as read.

FR-44 The system shall deliver notification by e-mail or SMS. *(Deferred.)*

4.8 Impact Reporting

Priority: Essential. The mechanism by which OBJ-2 is evaluated.

FR-45 The system shall compute and report servings redistributed, completed donations, donations lost to expiry, the unmatched rate, the waitlist size and waste diverted.

FR-46 The system shall compute median and mean time-to-match from the interval between a donation's submission timestamp and the response timestamp of its first accepted offer.

FR-47 The system shall compute every reported metric server-side from persisted timestamps; the client shall not derive any metric.

FR-48 The system shall report the first-offer acceptance rate and the mean donor-to-recipient distance. *(Deferred.)*

FR-49 The system shall collect a 1–5 clarity rating of the status timeline from both parties after completion. *(Deferred.)*

4.9 Organisation Verification

Priority: Deferred. In release 1.0 legitimacy is screened per donation (Section 4.3) rather than per organisation.

FR-50 The system shall allow a recipient organisation to upload registration documents and shall record a verification status of `PENDING`, `APPROVED` or `REJECTED`. *(Deferred.)*

FR-51 The system shall exclude an unverified organisation from matching. *(Deferred.)*

4.10 Unattended Operation

Priority: Deferred, and the highest-priority item for the next iteration. In release 1.0 expiry and re-matching occur only when an administrator triggers a cycle.

FR-52 The system shall expire overdue donations and offers automatically on a schedule, without administrator action. *(Deferred.)*

FR-53 The system shall surface a manual list of nearby organisations to the donor when automated matching finds none. *(Deferred.)*

4.11 Administrative Oversight

Priority: Deferred. In release 1.0 an administrator's view of the system is limited to the pending-review queue and the matching waitlist; there is no global listing of donations, matches and pickups and no account administration beyond a user's own profile.

FR-54 The system shall present an administrator with a filterable view of all donations, offers and pickups across every user, independent of review state. *(Deferred.)*

FR-55 The system shall allow an administrator to deactivate, reactivate or change the role of a user account, and shall record the acting administrator and the time of each such change. *(Deferred.)*

5 Other Nonfunctional Requirements

5.1 Performance Requirements

NFR-1 The median time-to-match shall be 20 minutes or less during the pilot window (OBJ-2).

NFR-2 Candidate retrieval shall use a spatial index so that it does not degrade linearly with the number of registered organisations.

NFR-3 A matching cycle over the pilot dataset shall complete within 5 seconds.

NFR-4 An interactive request shall complete within 2 seconds under pilot load.

NFR-5 The client shall remain usable on a low-to-mid range mobile device over a mobile data connection.

Note on NFR-1. The implemented workflow places an administrative review between submission and matching, so the measured interval includes administrator response time. The pilot must therefore either record review latency separately or measure from approval rather than from submission, and must state which definition it uses. This is recorded as TBD-1 in Appendix C.

5.2 Safety Requirements

NFR-6 No donation shall be matched before an administrator has approved it.

NFR-7 The system shall not assign a donation to an organisation whose declared dietary constraints exclude its food type.

NFR-8 The system shall not promise the same capacity to two donations simultaneously.

NFR-9 A donation reaching its deadline uncollected shall be recorded as expired and counted against the system's own performance, not silently discarded.

5.3 Security Requirements

NFR-10 All passwords shall be stored as one-way hashes using an adaptive hashing function.

NFR-11 Authorisation shall be enforced at the server for every operation; the client shall not be treated as a security boundary.

NFR-12 A user shall be able to read and modify only their own resources, except an administrator acting within their defined operations.

NFR-13 Personal contact details shall be disclosed only after an offer has been accepted.

NFR-14 Collected personal data shall be minimised to what the workflow requires.

NFR-15 The production deployment shall not use a development signing secret.

5.4 Software Quality Attributes

NFR-16 **Testability.** The matching service shall be testable in isolation, taking all state as input and returning a result, with no database dependency.

NFR-17 **Explainability.** Every allocation shall be reconstructable after the fact from stored score components and recorded exclusion reasons.

NFR-18 **Modifiability.** The allocation strategy shall be replaceable without modification to the API or the client.

NFR-19 **Auditability.** Every state transition shall be persisted with its timestamp and its actor.

NFR-20 **Usability.** A donor shall be able to complete a submission on one screen without scrolling past a single form.

NFR-21 **Reliability.** Staging and production availability shall be 99% or better during the pilot window.

NFR-22 **Interoperability.** The API contract shall be verifiable by an automated conformance check that fails the build on any breach.

5.5 Business Rules

BR-1 Only an administrator may approve or reject a donation.

BR-2 Only the owning donor may cancel a donation, and only before collection.

BR-3 Only the organisation an offer was made to may accept or reject it.

BR-4 Only the collecting organisation may record collection and confirm receipt.

BR-5 Acceptance is a property of an offer, not of a donation. A donation may be offered to several organisations in sequence.

BR-6 An organisation that rejects an offer is excluded from the immediate retry for that donation.

BR-7 A donation is never discarded for want of an eligible organisation; it is waitlisted and retried.

6 Other Requirements

6.1 Database Requirements

Persistence uses a relational database with a spatial extension. Locations are stored as geography points under SRID 4326 with GiST indexes, so that radius-constrained retrieval is evaluated in the query rather than in application code. The schema comprises ten tables: users, donors, recipient organisations, donations, matches, pickups, donation reviews, the matching waitlist, notifications and the geocoding cache.

6.2 Internationalisation

Release 1.0 is English-only, with distances in kilometres, quantities in servings and timestamps displayed in the viewer's local timezone. Multilingual support is out of scope.

A Glossary

Term	Definition
Donation	A listing of surplus food offered by a donor.
Match / Offer	A proposal of one donation to one organisation, carrying its scores and a response deadline.
Pickup	The record of a physical handover, with its own lifecycle.
Time-to-Match	Interval between a donation's submission and the response time of its first accepted offer.
Rescue rate	Proportion of posted donations reaching `COMPLETED` before their deadline.
Waitlist	Donations for which no organisation was eligible, retained for retry.
Capacity fit	Normalised measure of how well a donation's size suits an organisation's remaining capacity.
Urgency	Normalised measure of how little time remains before a donation expires.
Greedy assignment	Strategy processing donations in urgency order, committing each to its best available organisation.
Hungarian method	Classical algorithm solving the one-to-one assignment problem optimally over all open donations.
PostGIS	Spatial extension to PostgreSQL providing geography types and spatial indexes.

B Analysis Models

B.1 Donation State Machine

Figure 3 shows the permitted transitions. Each is validated centrally; an illegal transition is refused with HTTP 409 and has no partial effect. Acceptance does not appear as a donation state: it is a property of the offer (BR-5).

B.2 Matching Pipeline

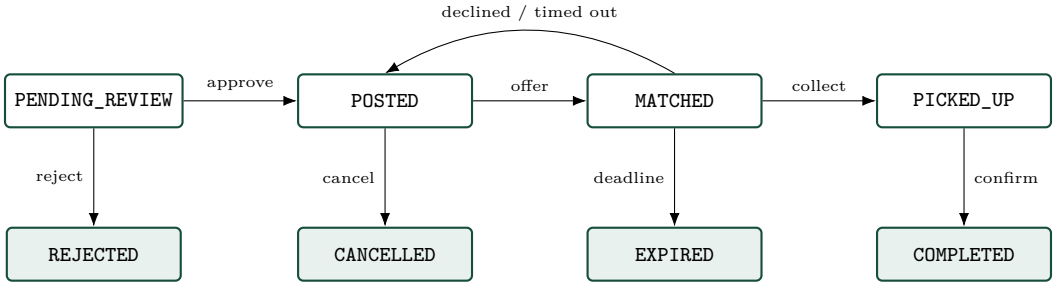


Figure 3: Donation lifecycle. The curved return from `MATCHED` to `POSTED` is the re-match path (FR-34, FR-35).

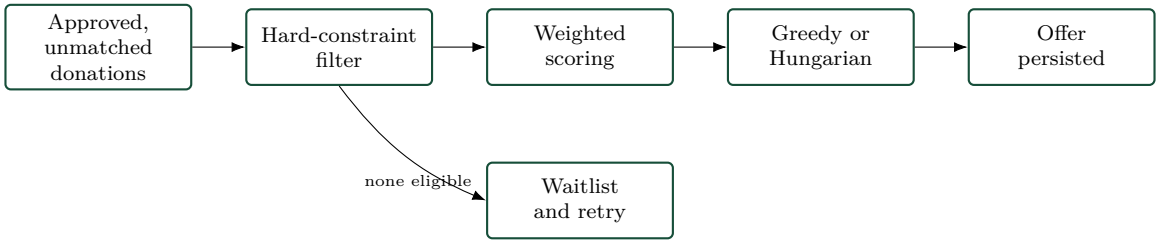


Figure 4: Filter, then rank. Eligibility is settled before any scoring, so an organisation failing a hard constraint never competes on score (FR-22).

C Open Issues

ID	Issue
TBD-1	Whether time-to-match is measured from submission or from administrative approval. Affects NFR-1 and the pilot's baseline comparison.
TBD-2	Whether a fairness term damping repeat allocation to the same organisation is added as a fourth scoring component.
TBD-3	The response window of 30 minutes is provisional and should be calibrated against observed organisation response times during the pilot.
TBD-4	Whether organisation verification (FR-50, FR-51) replaces or supplements per-donation review.